

Automated Dataset Generation & Evaluation Framework

Operational Excellence Initiative

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Using LLM-as-a-Judge for GenAI Chatbot Models

This framework automates dataset generation and evaluation of GenAI chatbot models using LLM-as-a-Judge methodology. By combining automated question-answer creation, human-in-the-loop validation, and AI-driven scoring, the process reduces manual effort, increases evaluation consistency, and accelerates model improvement cycles.

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Process Overview

1. Automated Q&A Generation - LLM creates diverse question-answer pairs from provided documents.
2. Human Review - Cleanup and validation to ensure accuracy and coverage.
3. Automated Evaluation - LLM compares model responses with ground truth using defined metrics.
4. Human Analysis - Review anomalies and adjust model strategies.
5. Dataset Reuse - Store validated Q&A for repeated testing and benchmarking.

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Process Flowchart

[Flowchart showing data input -> Q&A generation -> Human review -> Evaluation -> Metrics -> Analysis -> Improvement]

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Technical Details

- Language: Python
- Libraries: openai / transformers, evaluate, pandas, numpy, matplotlib, seaborn
- Dataset Format: JSON / CSV with question, ground_truth, model_response, metrics
- Evaluation: LLM-as-a-Judge compares responses against ground truth, calculates metrics, applies thresholds.

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Benefits & Time Savings

- Significant reduction in manual work from days to hours.
- Consistent scoring across multiple iterations.
- Scalable to thousands of test cases.
- Reusable datasets enable fast regression testing.
- Clear and auditable evaluation process.

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Reliability Safeguards

1. Human validation at critical points.
2. Ground truth-based scoring.
3. Clear metric definitions and thresholds.
4. Multiple evaluations for consensus.
5. Full audit logs of judge reasoning.

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Sample Metrics Dashboard

[Sample bar chart and table visualizing model performance on metrics: correctness, relevance, faithfulness, context relevance]